



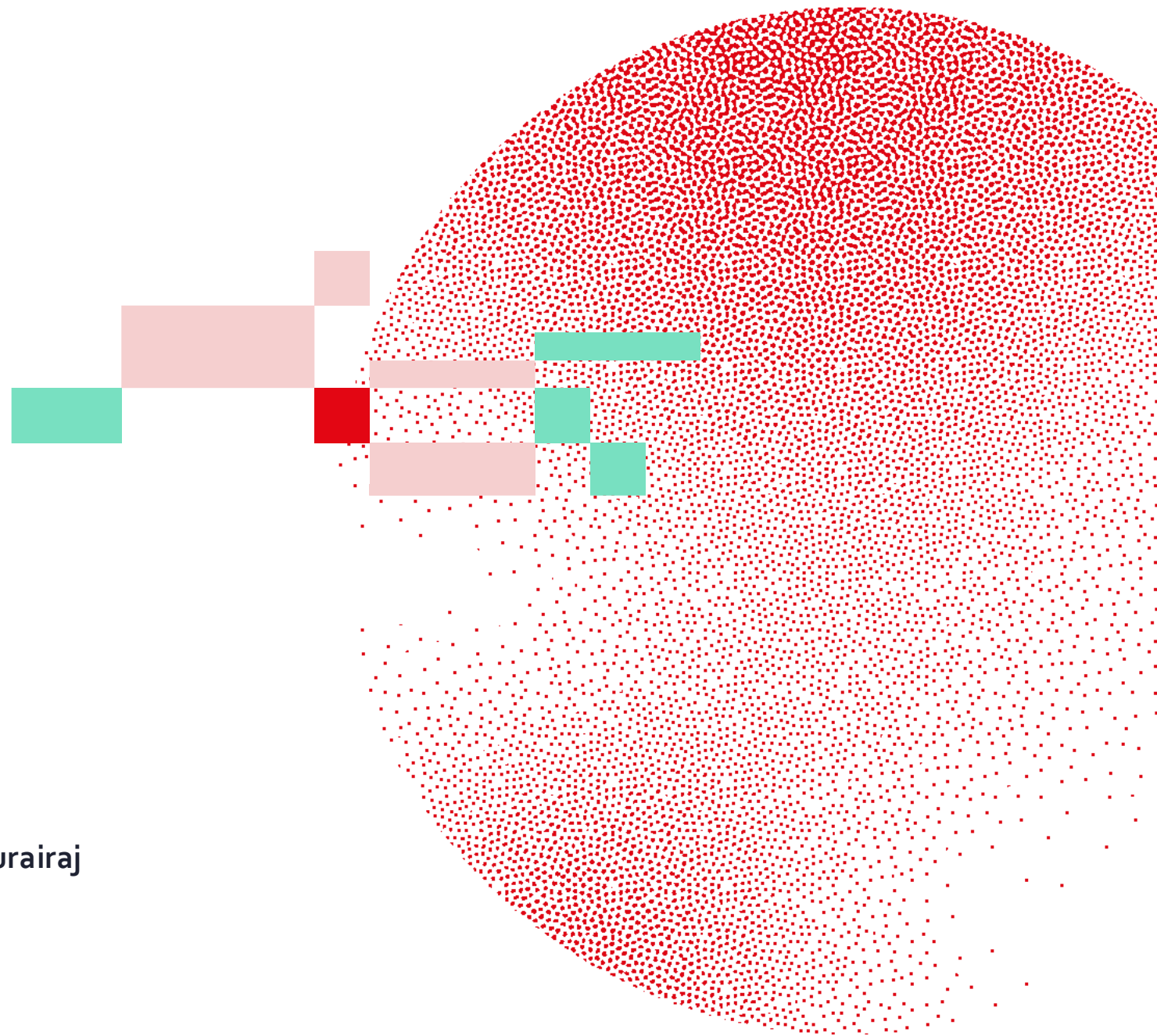
Swiss Institute of
Bioinformatics

DAY 2, PART 4

Bonus methods

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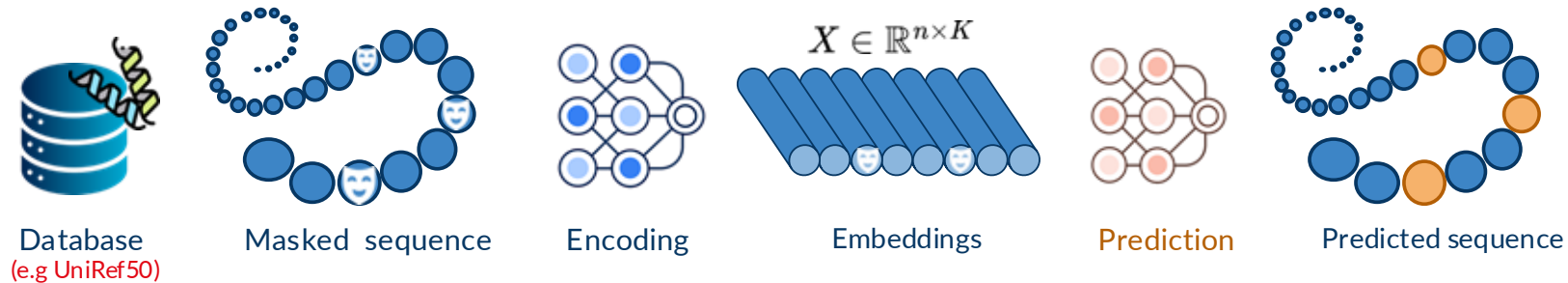
23-24 June 2026



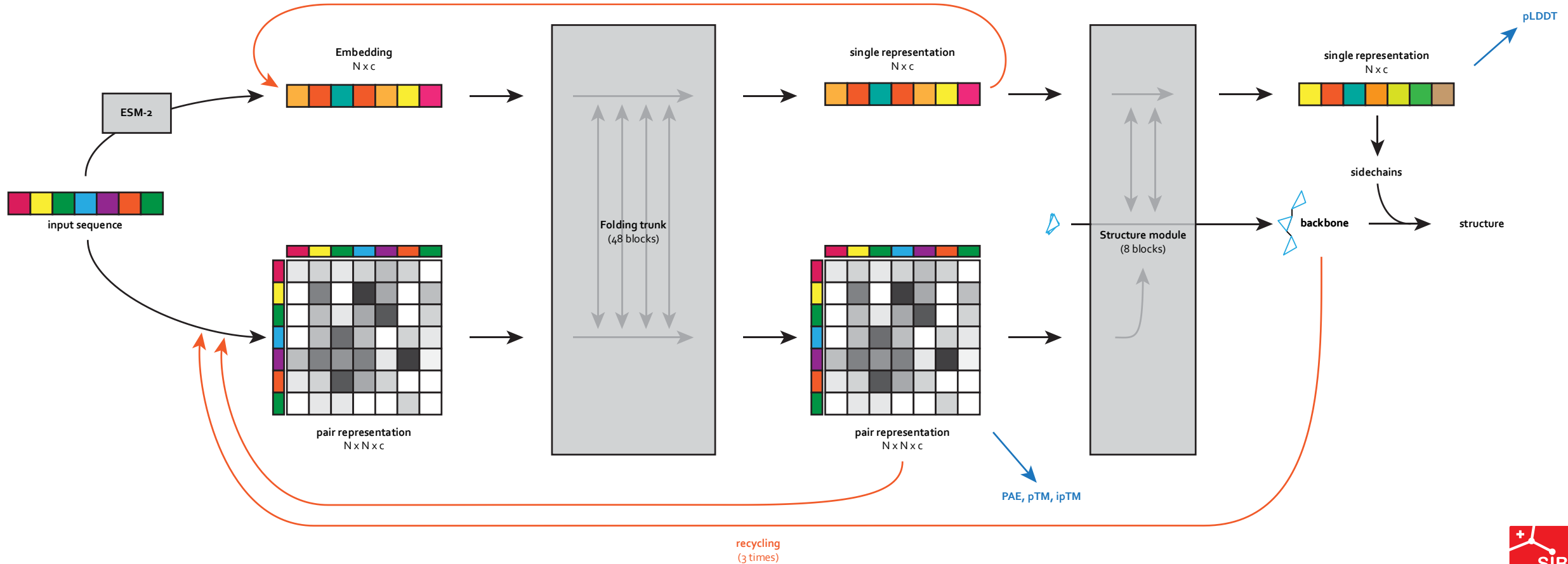


ESM-2

- Protein language model (pLM)
- Trained on 65 million sequences
(compared to billions of sequences AlphaFold2 has access to when constructing an MSA)



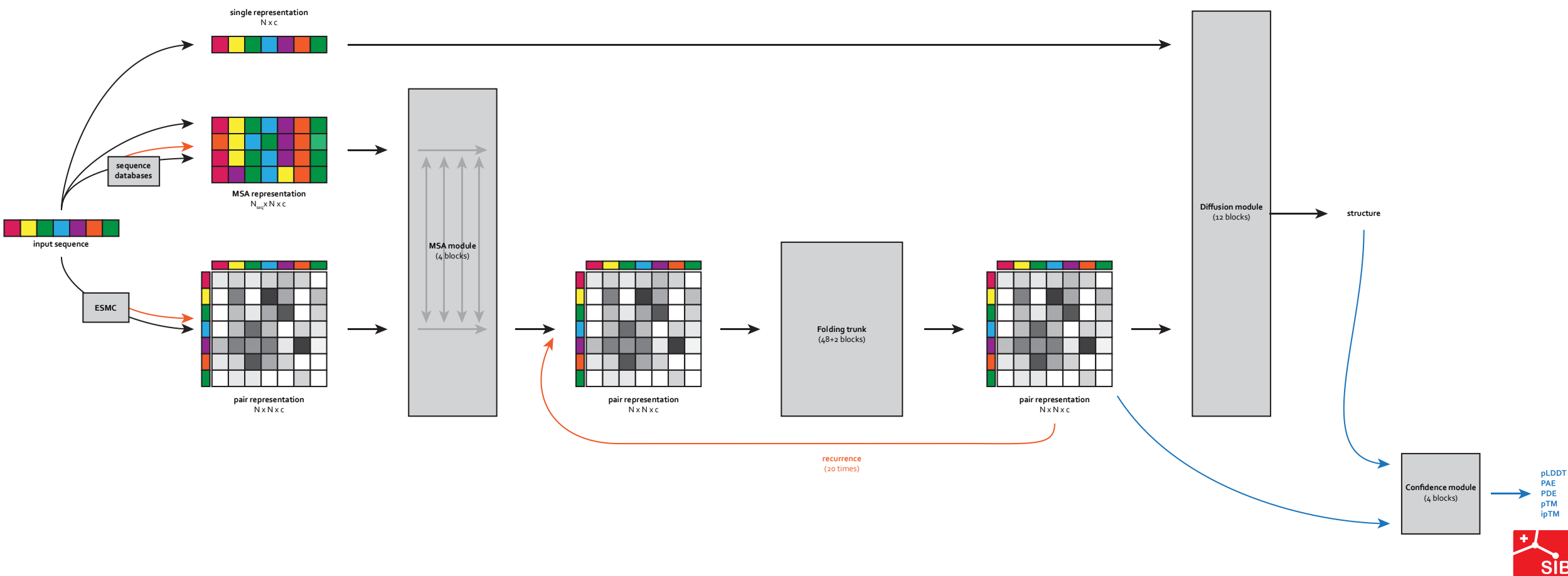
- Similar to AlphaFold2, but no templates, no MSA → fast
- Doesn't perform as well as AlphaFold2



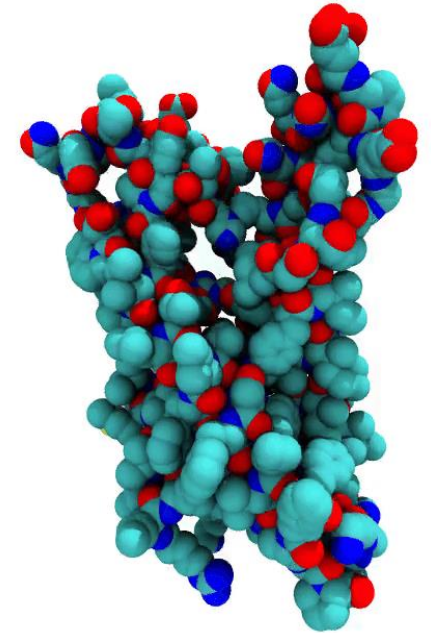


ESMC, ESMFold2

- Trained on 2.8 billion sequences
- Better at antibodies and protein-protein interactions; there is also a fast version



- AlphaFold is trained to predict only one conformation (crystal structure)
- Proteins are very dynamic in their native environment (solution, temperature, concentration)
- Methods to simulate protein dynamics:
 - MD – Molecular dynamics (physics based, takes a lot of time to simulate a few microseconds)





- Similar architecture to AlphaFold2
 - Replaces the structure module with a denoising diffusion model
- Trained to predict ensembles of conformations
 - Static structures, MD simulations, equilibrium distributions

